



QuantaLayer Research Note #1

Preparing Solana for the Post-Quantum Era

State of the Art, Limits, and the Case for a Readiness Layer

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QuantaLayer Research Note. Not a security audit. Not legal advice. Not investment advice. QES is a migration criticality score, not a hack probability score. QuantaLayer Vault is experimental and not mainnet-ready. Superteam Balkan membership is a personal affiliation and does not imply endorsement.

Abstract

Solana's current account and transaction model relies heavily on Ed25519 signatures. A cryptographically relevant quantum computer capable of running Shor's algorithm at scale would invalidate the security assumptions behind Ed25519, although no public system demonstrates that capability today. This research note reviews the post-quantum readiness problem for Solana without presenting it as an immediate break. It summarizes the relevant cryptographic background, the NIST post-quantum standards, the non-final status of FIPS 206/FN-DSA, and Solana-specific constraints such as address-as-public-key accounts, Program Derived Addresses, transaction identifiers, fee payers, upgrade authorities, token authorities, stake and vote accounts, validator identities, multisigs and governance keys. It then reviews current ecosystem work by Solana Foundation, Anza, Firedancer, Jump Crypto, Project Eleven, Helius and Blueshift. QuantaLayer is positioned as an application and analytics layer for post-quantum readiness: QuantaLayer Scan measures migration criticality through QES and data confidence through QCI; Authority Exposure prioritizes control keys for protocols and institutions; QuantaLayer Vault remains an experimental opt-in component; and QuantaLayer Notary anchors hashes of hybrid off-chain certificates. QuantaLayer does not replace protocol-level migration.

Keywords: post-quantum cryptography; Solana; Ed25519; Falcon; FN-DSA; ML-DSA; WOTS; crypto-agility; migration criticality; QuantaLayer.

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1. Introduction

The security of public blockchains depends on the assumption that only the holder of a private key can authorize a valid state transition. On Solana, this assumption is primarily instantiated with Ed25519 signatures. A cryptographically relevant quantum computer (CRQC) capable of running Shor's algorithm at cryptographic scale would break the elliptic-curve discrete logarithm assumptions behind Ed25519 and related schemes [1, 2].

That is not the same as saying Solana is practically broken today. No public quantum computer demonstrates the ability to recover Ed25519 private keys. The engineering issue is different: post-quantum migration is a long-lead, multi-actor change affecting accounts, programs, wallets, validators, governance processes, custody workflows and compliance surfaces. The relevant question is not when to generate fear, but how to build inventory, measurement, prioritization and experimentation before a rushed migration becomes necessary.

Solana has several characteristics that make this problem different from Bitcoin or Ethereum. For keypair-controlled accounts, the address is the Ed25519 public key itself. Program Derived Addresses (PDAs) are off-curve addresses without private keys, but their safety depends on the programs and authorities that control them. The fee payer, the first signature used as the transaction identifier, versioned transactions, Address Lookup Tables, compute limits and account locking all shape the migration path [3, 4, 5, 6, 7, 8].

The Solana ecosystem is already studying this transition. Solana Foundation has published a quantum-readiness roadmap; Anza and Firedancer have explored Falcon/FN-DSA; Jump Crypto has analyzed migration paths; Project Eleven has reported a Solana testnet effort; and Blueshift/Dean Little published hash-based vault work before QuantaLayer [9, 10, 11, 12, 13]. QuantaLayer is designed to complement this work through readiness measurement, reporting and bounded experimentation, not to replace protocol research.

Key takeaways

1. Solana is not practically broken today.
2. Post-quantum migration affects more than wallet signatures.
3. Authorities and governance keys are critical migration targets.
4. QuantaLayer provides measurement and readiness, not protocol migration.
5. Vault and Notary remain bounded, opt-in components that require independent review before production claims.

2. Methodology

This note uses a conservative source hierarchy. Primary standards and official technical documentation are treated as authoritative for protocol or standard status. Ecosystem announcements are treated as evidence of intent or research direction, not as proof of production deployment. Press sources are used only to document reported claims, never as benchmark evidence.

Claims are stated with three constraints. First, CRQC risk is conditional: it depends on future hardware and error correction, not on an observed present capability. Second, proposal status is not adoption: a GitHub pull request, a prototype or a testnet is not a mainnet protocol decision. Third, readiness metrics are operational triage tools, not cryptographic proofs.

Table 1: Source hierarchy used in this research note

Tier	Examples	Use in the note
Standards and RFCs	NIST FIPS 203/204/205, RFC 8032, RFC 8785	Cryptographic definitions and final standard status.
Official protocol documentation	Solana documentation, Anza documentation, Metaplex documentation	Account model, transactions, token and program surfaces.
Primary ecosystem sources	Solana Foundation, Jump Crypto, Helius, Blueshift repositories	Migration paths, proposals, implementation signals.
Research papers and whitepapers	Shor, Grover, Google Quantum AI, Falcon specification	Threat model and cryptographic trade-offs.
Press and secondary reports	CoinDesk, Unchained	Reported context only; not benchmark proof.

3. Cryptographic Background

3.1. Quantum algorithms and classical assumptions

Shor’s algorithm threatens public-key schemes based on integer factorization and discrete logarithms, including elliptic-curve signatures [1]. Grover’s algorithm gives a quadratic speedup for unstructured search, which affects symmetric security margins and hash preimage costs but does not invalidate hash functions in the same structural way [14]. Hash-based signatures therefore remain an important post-quantum family when parameterized conservatively.

Solana uses Ed25519 signatures for ordinary transaction authorization [2]. Under a future CRQC, exposed Ed25519 public keys and signatures become migration-relevant artifacts. The distinction is important: visible public-key material is an exposure surface, but exposure is not itself an exploit.

3.2. NIST standards and Falcon/FN-DSA status

NIST finalized FIPS 203 (ML-KEM), FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA) in 2024 [15, 16, 17]. FIPS 206 / FN-DSA / Falcon is not final at the time of writing. NIST material identifies Falcon/FN-DSA as a standardization track, not as a completed FIPS [18, 19].

Table 2: Post-quantum standards status relevant to this note

Reference	Primitive	Status used in this note
FIPS 203	ML-KEM	Final NIST KEM standard.
FIPS 204	ML-DSA	Final NIST lattice-based signature standard.
FIPS 205	SLH-DSA	Final NIST stateless hash-based signature standard.
FIPS 206	FN-DSA / Falcon	Not final; parameters, encodings and implementation guidance may evolve.

Falcon is attractive for Solana because signatures are compact relative to ML-DSA and SLH-DSA. That benefit comes with implementation complexity: Gaussian sampling, constant-time behavior and side-channel resistance are harder to engineer than for simpler APIs. A Solana migration path involving Falcon should therefore be evaluated through audited implementations, reproducible benchmarks and governance review, not by signature size alone [20, 11].

Table 3: Indicative signature and public-key sizes

Scheme	Public key	Signature	Comment
Ed25519	32 B	64 B	Current Solana transaction signature scheme.
ML-DSA-65	1,952 B	3,309 B	Final NIST signature option, large for Solana transactions.
SLH-DSA	parameter-dependent	much larger	Stateless hash-based standard; useful for diversity, costly for on-chain use.
Falcon-512	about 897 B	about 666 B average	Compact, but FIPS 206 is not final.
WOTS/WOTS+	parameter-dependent	one-time, hash-based	Useful for opt-in vault designs; requires strict key-use discipline.

4. Solana-Specific Constraints

4.1. Accounts, addresses and signatures

For keypair-controlled accounts, a Solana address is the Ed25519 public key. Helius notes that post-quantum public keys are too large to become 32-byte addresses directly; a plausible migration would derive addresses from a hash of the post-quantum public key and a scheme identifier, with an off-curve or collision-avoidance rule analogous in spirit to PDA derivation [21]. This would affect wallet UX, transaction formats, account discovery and signing semantics.

The current transaction identifier depends on the first Ed25519 signature. A post-quantum transaction model would more naturally use a hash of the message payload or a redesigned identifier rule, because post-quantum signatures are larger and scheme-dependent [21, 5]. Fee payer semantics and compute budget constraints also matter: verification cost is not an abstract benchmark but a runtime resource allocation problem [6].

4.2. Program and authority surfaces

Solana's risk surface is not limited to user wallets. Upgradeable programs use ProgramData accounts and upgrade authorities through the BPFUpgradeableLoader model [22]. Token mints may have mint, freeze or close authorities. Metaplex assets have update authorities. Stake accounts have withdraw authorities. Vote accounts and validator identity keys are operationally critical [23, 24, 25, 26, 27, 28].

Table 4: Solana surfaces that affect post-quantum readiness

Surface	Examples	Readiness implication
Account model	System Program accounts, keypair-controlled accounts, PDAs	Addresses, ownership and signer rules are migration-critical.
Transactions	fee payer, first signature as transaction id, compute budget, versioned transactions, Address Lookup Tables	Larger signatures and different identifiers require runtime and wallet changes.
Programs	BPFUpgradeableLoader, Program-Data, upgrade authority	Protocol control can be concentrated in Ed25519 authorities.
Tokens	SPL Token, Token-2022, mint/freeze/close authorities	Asset migration depends on authorities, not only holders.
NFT/cNFT	Metaplex Token Metadata, Bubblegum compressed NFTs	Update authority and indexing quality affect exposure inventory.
Staking and voting	stake withdraw authority, vote authority, validator identity	Validator and institutional operations require special migration planning.
Governance	Squads multisig, Realms governance, treasury authorities	Multi-party control keys are high-priority B2B targets.

5. State of the Art in Solana Post-Quantum Readiness

Solana Foundation describes quantum readiness as a long-term engineering problem and reports convergence by Anza and Firedancer toward Falcon/FN-DSA as a compact candidate for high-throughput signatures [9, 10]. Jump Crypto similarly treats Falcon as a leading candidate while noting that different components may evolve independently [11].

SIMD-0461 existed as a proposal for Falcon-512 verification on-chain. It was initially formulated as a precompile and later refactored as a syscall under the title “SIMD-0461: enabling falcon signature verification as a syscall”. The official pull request #461 in ‘solana-foundation/solana-improvement-documents’ is closed; it was closed on June 17, 2026. The author indicated that the effort was paused, mentioned a Dean Little Falcon implementation in SBF with reasonable compute cost, and wrote that it was closed for now and could be reopened when there is more demand [29]. It is a meaningful research signal, but it is not active, merged, adopted or a confirmed protocol trajectory.

Google Quantum AI’s 2026 whitepaper distinguishes attacks against public keys already exposed at rest from attacks during the spend window. It gives indicative timing comparisons of roughly 400 ms for Solana, 12 seconds for Ethereum and 10 minutes for Bitcoin, and includes a “Post-Quantum Blockchains” discussion covering Solana Winternitz Vault, Algorand, XRP Ledger, QRL, Mochimo and Abelian [30]. The same paper does not demonstrate a currently operational CRQC.

Blueshift and Dean Little deserve explicit priority for Solana hash-based vault work. The Solana Winternitz Vault repository describes WOTS with truncated Keccak-256 and 224-bit preimage resistance, while Blueshift’s Winterwallet article discusses a related but not automatically identical WOTS design and parameterization [13, 31, 32]. QuantaLayer Vault does not claim to introduce a new primitive.

Table 5: Selected ecosystem work and how it is treated

Actor	Contribution	Interpretation in this note
Solana Foundation	Quantum-readiness roadmap	Official ecosystem direction, not final migration deployment.
Anza / Firedancer / Jump	Falcon/FN-DSA research and migration paths	Strong technical signal; production path still depends on standardization, audits and governance.
SIMD-0461	Falcon verification as a syscall	Closed proposal; research signal only.
Project Eleven	Threat assessment and reported testnet work	Useful context; public material is not a reproducible benchmark protocol.
Helius	Solana-specific migration analysis	Key source for address, transaction and authority implications.
Blueshift / Dean Little	Winternitz Vault and Winterwallet work	Prior Solana hash-based vault work that QuantaLayer must acknowledge.

6. Current Limits and Open Problems

Post-quantum readiness for Solana remains constrained by incomplete standardization, runtime costs, governance sequencing, wallet UX and public benchmark gaps. These limits are not blockers to research, but they prevent responsible claims of production readiness.

Table 6: Current limits that must remain visible

Limit	Why it matters	Publication-safe statement
FIPS 206 is not final	Falcon parameters and encodings may evolve	Treat Falcon as promising but provisional.
No adopted Falcon syscall	SIMD-0461 is closed	Do not describe Falcon verification as active protocol work.
Transaction and compute limits	PQ signatures are larger and verification can be costly	Benchmark with real Solana runtime constraints.
Persistent authorities	Protocol and treasury keys may control more value than wallets	Prioritize Authority Exposure.
Indexing gaps	DeFi, NFT/cNFT and price data may be incomplete	Couple QES with QCI and with warnings.
Benchmark opacity	Public reports may lack scripts, logs and distributions	Require reproducible protocols before performance claims.

ANSSI guidance supports inventory, crypto-agility and hybridization for regulated security-product contexts [33, 34, 35]. A 2027 product-qualification milestone has been reported in specialized press, but without a primary doctrine located for this note it should not be presented as a direct ANSSI rule for public blockchains or as a commercial argument for QuantaLayer.

7. QuantaLayer Scan: QES and QCI

QuantaLayer Scan is a read-only scanner. It never requests private keys, seed phrases or transaction signatures for a public address scan. Its purpose is to estimate migration criticality and data confidence from observable public data.

Let a be a Solana address, $O(a)$ the set of observable scoring factors for that scan, $f_i(a) \in [0, 1]$ a normalized factor value, w_i the nominal factor weight and w'_i the renormalized weight over the observable set. QuantaLayer defines:

$$\text{QES}(a) = \text{round} \left(100 \sum_{i \in O(a)} w'_i f_i(a) \right), \quad w'_i = \frac{w_i}{\sum_{j \in O(a)} w_j}.$$

QES is a migration criticality score, not a probability that an address will be hacked. If a factor is not observable, the remaining weights are renormalized and QCI must decrease.

Table 7: QES factors

Factor	Nominal weight	Interpretation
Observable asset value to migrate	0.35	Estimated migration exposure value, not financial VaR.
Staked or locked assets	0.20	Operational friction and time to move or redelegate.
Observable concentration	0.15	Lack of compartmentalization when visible.
Observable age	0.10	First indexed activity, first known signature or first indexer appearance; not guaranteed account creation date.
Significant active token accounts	0.10	Migration complexity after dust/spam filtering.
Recent activity	0.10	Likelihood that the account can be reached or migrated proactively.

The token-account factor must be resistant to dusting. QuantaLayer Scan should apply a minimum value threshold, ignore or underweight spam tokens, distinguish active token accounts from unsolicited dust, weight meaningful assets more heavily and reduce QCI when pollution prevents reliable interpretation.

For confidence, let $c_k(a) \in [0, 1]$ be a confidence component and v_k its weight:

$$\text{QCI}(a) = \text{round} \left(100 \sum_k v_k c_k(a) \right).$$

Table 8: QCI dimensions and display rule

Dimension	Weight	Method
Resolved prices	0.30	Share of value covered by reliable prices.
RPC/DAS responses	0.20	Complete pagination and valid schemas.
DeFi positions	0.15	LP, lending and perps decoded where relevant.
NFT/cNFT	0.10	Indexing and valuation coverage.
Stake accounts	0.10	Amounts and authorities resolved.
Data freshness	0.10	Cache age and TTL.
Account classification	0.05	System, PDA, multisig, program or other class.

Table 9: QCI controls how QES is displayed

QCI	User display
≥ 80	QES displayed normally.
60–79	QES displayed with warning.
40–59	QES displayed as a fragile estimate.
< 40	No A/B/C/D/E grade; show insufficient data.

8. Authority Exposure Module

Authority Exposure is the most important B2B module because protocol, DAO and institutional risk is often concentrated in control keys rather than ordinary wallet balances. A compromised upgrade authority can change a program. A compromised mint authority can affect token supply. A compromised treasury or governance authority can redirect assets even if individual user wallets are not directly affected.

The module should produce an inventory and migration criticality analysis for authorities. It is not a security audit and does not replace manual review of program logic, governance processes or custody controls.

Table 10: Authority Exposure coverage

Authority type	Examples	Why it matters
Program control	upgrade authority, ProgramData account, program deployer, BP-FUpgradeableLoader	Can modify executable logic or governance-controlled protocol behavior.
Token control	mint authority, freeze authority, close authority, Token-2022 authorities	Can affect supply, transferability or account lifecycle.
Metadata and assets	metadata update authority, Bubblegum/cNFT authorities	Can affect NFT/cNFT integrity and presentation.
Staking and validation	stake withdraw authority, vote authority, validator identity	Operationally critical for validators and institutions.
Governance and treasury	DAO governance keys, Realms governance, Squads multisig owners, protocol treasury authorities	Often controls high-value or high-impact actions.

9. QuantaLayer Vault

QuantaLayer Vault is an experimental, non-custodial, opt-in vault. It protects only assets actually deposited into it. It does not protect the Solana account model, validators, consensus, fee payers or external authorities. QuantaLayer Vault must not be used for real funds until public devnet testing, independent audits, capped release rules and recovery documentation are published.

QuantaLayer recognizes the priority of Dean Little / Blueshift and the Solana Winternitz Vault. QuantaLayer Vault does not claim to introduce a new primitive; it proposes a benchmarked, audited and UX-integrated experimental path linked to QES/QCI.

Table 11: Distinguishing Solana Winternitz Vault, Winterwallet and QuantaLayer Vault

Element	Source	Parameters
Solana Winternitz Vault GitHub	Blueshift repository	WOTS plus truncated Keccak-256 with 224-bit preimage-resistance target.
Winterwallet	Blueshift article	WOTS, hash and truncation parameters as documented by Blueshift; not assumed identical to the repository.
QuantaLayer Vault	QuantaLayer	Not fixed; benchmark and audit are mandatory before any release claims.

A refund address can reintroduce residual Ed25519 exposure. QuantaLayer Vault must display the QES/QCI of the refund address, recommend refunding to a new vault or post-quantum-compatible strategy where appropriate, and test the scenario where a refund address is highly critical.

No mainnet QuantaLayer Vault should exist without public devnet testing, Bankrun and LiteSVM coverage, instruction fuzzing, property-based tests, anti-replay and malleability tests, weak-RNG simulations, benchmark comparison with a precise Blueshift commit, two independent audits, a bug bounty, caps, recovery documentation, incident monitoring, legal review and a clear upgrade-authority strategy.

10. QuantaLayer Notary

QuantaLayer Notary is a hybrid document-integrity and anchoring design. ML-DSA-65 is used off-chain in the certificate. The on-chain transaction is not the certificate itself; it is an anchor to a certificate hash.

Table 12: QuantaLayer Notary processing flow

Step	Artifact	Location / role
Document	final file	issuer side
Canonicalization	deterministic representation	off-chain
Document hash	SHA-256	signed payload
Signatures	Ed25519 + ML-DSA-65	off-chain certificate
Certificate hash	SHA-256	on-chain anchor
Solana anchor	transaction / slot	public proof

The signed message must be domain-separated, versioned and canonical:

```
signature_message =
  domain_separator ||
  schema_version ||
  canonical(
    document_hash,
    issuer_id,
    issuer_key_id,
    issued_at,
    policy_id,
    revocation_registry,
    anchor_policy,
    chain_id,
    cluster_id,
    canonicalization_profile
  )
```

The public certificate must not contain personal data about the subject of the document. Issuer metadata must be minimized; if it identifies a natural person, it must be treated as personal data. JSON canonicalization should use RFC 8785/JCS or an internal versioned ‘json-c14n-v1’ profile [36]. For PDF, the hash should cover the final bytes, metadata must be frozen before signature, no implicit normalization is applied and any modification changes the hash.

QuantaLayer Notary is neither a qualified electronic signature nor a qualified timestamping service under eIDAS. Integrity evidence is not evidence that the content is true. Relative proof of existence is not a qualified timestamp.

Table 13: Minimum key registry fields for QuantaLayer Notary

Field	Purpose
issuer_id	Stable issuer identity or registry identifier.
issuer_key_id	Versioned key identifier used in signed messages.
alg	Algorithm identifier, for example ML-DSA-65.
public_key	Complete public key, not only a fingerprint.
fingerprint	Stable digest used by certificates and offline verifiers.
valid_from / valid_until	Validity window.
status	active, revoked or expired.
rotation_history	Previous and successor key relationships.
registry_anchor	Hash or transaction anchoring the registry state.
compromise_policy	Revocation and disclosure procedure after compromise.

11. Abuse Risks and Responsible Design

Readiness tools can be misused. A public score can become a phishing pretext, a targeting aid or a false safety badge. QuantaLayer must therefore design for conservative display, rate limits, anti-scraping controls and claim discipline.

Table 14: Abuse risks and mitigations

Risk	Failure mode	Mitigation
Targeting lists	Scanning third-party addresses to rank valuable accounts	Rate limits, aggregate-only public reporting, no nominative leaderboard.
Phishing	Fake reports or urgent migration prompts	Clear read-only messaging; never request seed phrases; signed official reports.
False precision	Treating QES as a hack probability	Always display QCI and explanations; no grade below QCI 40.
Vault overclaim	Users treat experimental vaults as audited custody	Mainnet prohibition until audits, caps and recovery documentation.
Notary overclaim	Integrity proof mistaken for legal qualification	Explicit eIDAS and truth-of-content limitations.

12. Roadmap and Open-Source Strategy

Table 15: QuantaLayer roadmap

Horizon	Deliverables	Conditions
2026 Q3	QuantaLayer Scan, QES/QCI, documentation, initial aggregate reports	Versioned constants, cited sources, limitations displayed.
2026 Q4–2027 H1	Authority Exposure, Vault devnet, benchmark protocol	Bankrun, LiteSVM, fuzzing, Blueshift comparison, legal review.
2027+	Notary B2B, periodic reports, capped Vault only if conditions are met	Independent audits, bug bounty, recovery documentation, monitoring.

QuantaLayer will publish at minimum the QES/QCI specification, versioned weights, invariants, synthetic test sets and reproducible examples. Publication of the full indexing, scoring and valuation pipeline will depend on open-source strategy, anti-abuse design, security considerations and intellectual property protection.

13. Discussion

Post-quantum readiness is a sequencing problem. Protocol teams must decide signature schemes, runtime rules and migration governance. Wallets must implement new signing flows. Institutions must inventory authorities and custody policies. Users must understand which assets and accounts require action. No single application layer can solve the protocol-level migration.

QuantaLayer’s defensible contribution is therefore narrower and more useful: make exposure visible, make confidence explicit, prioritize authorities and publish reproducible benchmarks. This framing also limits the product’s claims. QuantaLayer can help decide where migration work should start. It cannot certify that an account is safe, predict Q-Day, replace audits or substitute for a future Solana protocol migration.

14. Conclusion

Solana’s post-quantum transition is not a one-line cryptographic upgrade. It touches account semantics, address formats, transaction identifiers, fee payers, compute budgets, upgrade authorities, token authorities, validator operations and governance controls. Existing ecosystem work has already iden-

tified serious paths: Falcon/FN-DSA research by Solana client teams, analysis by Jump Crypto and Helius, Project Eleven’s reported testnet work, and Blueshift’s prior Winternitz-based vault designs.

QuantaLayer should be evaluated as a readiness layer within that broader ecosystem. QuantaLayer Scan measures migration criticality with QES and data confidence with QCI. Authority Exposure extends the model to protocol and institutional control keys. QuantaLayer Vault remains experimental and opt-in. QuantaLayer Notary anchors hashes of hybrid off-chain certificates. The value of the project depends on measurement discipline, careful limits, transparent benchmarks and refusal to overclaim.

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A. Claims Allowed and Prohibited

Table 16: Publication claim matrix

Allowed	Wording
Measurement	QuantaLayer measures migration criticality.
Prioritization	QuantaLayer helps prioritize post-quantum readiness.
Vault status	QuantaLayer Vault is experimental.
Authority Exposure	Authority Exposure is an inventory and criticality module.

Table 17: Prohibited publication claims

Do not claim	Reason
Protocol-protection claim for QuantaLayer.	Protocol-level security is outside the application layer.
Quantum-safe outcome claim for Solana through QuantaLayer.	This would overstate scope and timing.
Hack-prediction claim for QES.	QES is a migration criticality score.
Accuracy-guarantee claim for QCI.	QCI measures data completeness and confidence.
Production-safety claim for Vault before audits.	Vault remains experimental until independent audits are published and release caps are enforced.
Qualified-trust-service claim for Notary under eIDAS.	Notary is an integrity and anchoring design, not a qualified trust service.
Final-standard claim for FIPS 206.	FIPS 206 is not final at publication time.
Protocol-adoption claim for SIMD-0461.	The pull request is closed and not adopted.

B. QuantaLayer Vault Benchmark Protocol

B.1. Environment to publish

Every benchmark report must publish: QuantaLayer Vault commit, Solana Winternitz Vault commit, Anchor version, Solana toolchain, Bankrun version, LiteSVM version, hardware, hash function, WOTS/WOTS+ parameters, truncation, instruction size, transaction size, compute units, fees, localnet/devnet cluster, feature flags and upgrade-authority configuration.

Table 18: Benchmark metrics

Metric	Required reporting
Instruction and transaction size	bytes, including failure cases for transaction-too-large errors.
Compute units and fees	CU distribution and lamports by instruction.
Latency	p50, p95 and p99; no average alone.
Reliability	localnet and devnet success rates.
State size and lifecycle	initialized, rotated, interrupted, closed and partially initialized states.
Comparison	Same scenario against a precise Blueshift repository commit.

B.2. Required test scenarios

- transaction too large;
- compute exceeded;
- anti-replay;
- malleability;
- reused signature;
- wrong vault;
- wrong message;
- stale slot or nonce;

- weak RNG simulation;
- interrupted rotation;
- double close;
- partially initialized state;
- upgrade authority;
- refund address vulnerable;
- refund to new vault.

Benchmark publication requires distributions, scripts, logs, CSV/JSON raw data and reproduction instructions.

C. QuantaLayer Notary Certificate Specification

C.1. Minimal certificate

The following JSON is a minimal illustrative certificate schema. Production schemas must be versioned and validated against a formal JSON schema.

```
{
  "schema": "quantalayer-notary-v0.1",
  "domain_separator": "quantalayer-notary",
  "document": {
    "canonicalization": "pdf-bytes-v1",
    "hash_alg": "SHA-256",
    "hash": "base64url..."
  },
  "issuer": {
    "issuer_id": "did-or-domain-or-registry-id",
    "issuer_key_id": "key-2026-01",
    "policy_id": "quantalayer-policy-v0.1"
  },
  "issued_at": "2026-07-05T12:00:00Z",
  "revocation": {
    "registry": "https-or-solana-registry",
    "status": "active"
  },
  "anchor_policy": {
    "chain_id": "solana",
    "cluster_id": "mainnet-beta",
    "min_confirmations": 32
  },
  "classical_signature": {
    "alg": "Ed25519",
    "public_key": "base58...",
    "signature": "base64url..."
  },
  "post_quantum_signature": {
    "alg": "ML-DSA-65",
    "public_key_fingerprint": "base64url...",
    "signature": "base64url..."
  },
  "anchor": {
    "transaction": "signature...",

```


4. It does not treat Project Eleven public claims as reproducible performance evidence.
5. It does not assume that Winterwallet and the Solana Winternitz Vault repository use identical parameters.
6. It does not treat a QES grade as valid when QCI is below 40.
7. It does not replace protocol audits, custody reviews or legal analysis.

F. Release Integrity Protocol

The public release is produced from the LaTeX source and bibliography by the repository build script. The script compiles the PDF, cleans auxiliary files, generates a detached SHA-256 checksum manifest, copies the PDF and checksum into the public article directory, and creates a linearized PDF when ‘qpdf’ is available.

Table 19: Publication release artifacts

Artifact	Release role
Canonical PDF	Public document linked from the title page.
Detached ‘.pdf.sha256’ file	Verifies the final PDF without embedding a self-referential hash inside the document.
Compile report	Records build command, PDF metadata, embedded fonts, checksum verification, ‘qpdf’ validation and rendered-page inspection.